Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
Mid Semester Examination (MST)

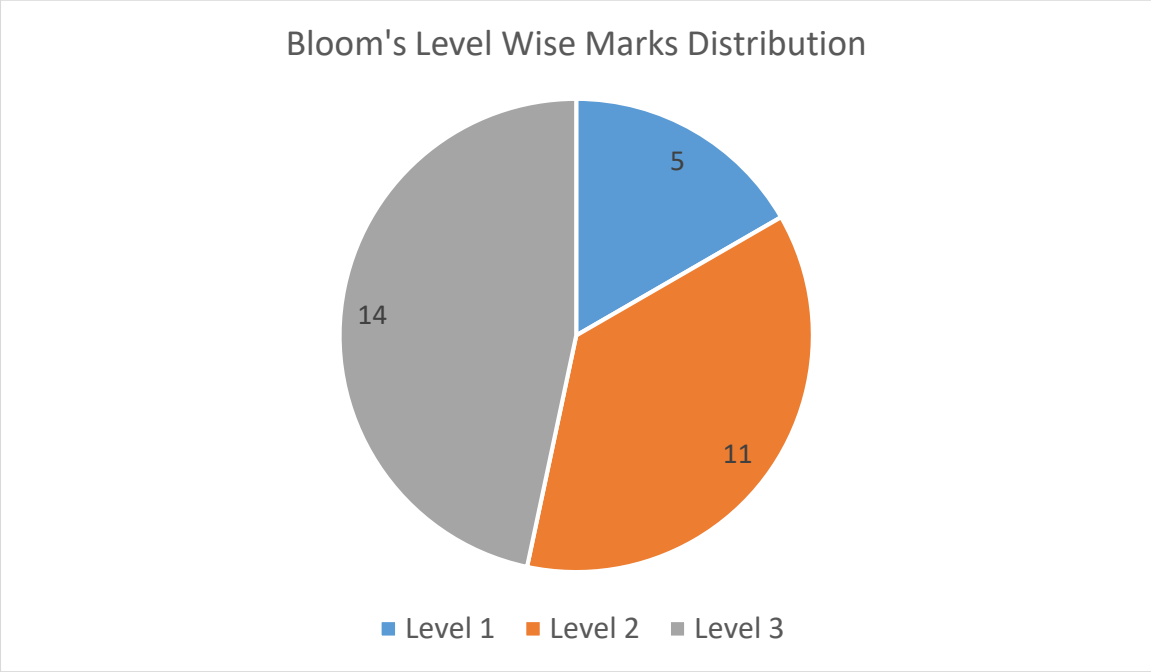
B.E (6th Semester) **Course Code:UCS677**
23rd March, 2024 **Course Name: Data Engineering**
Maximum Marks: 30 **Faculty: Dr. Sumana Maiti, Dr. Mandeep Kaur**

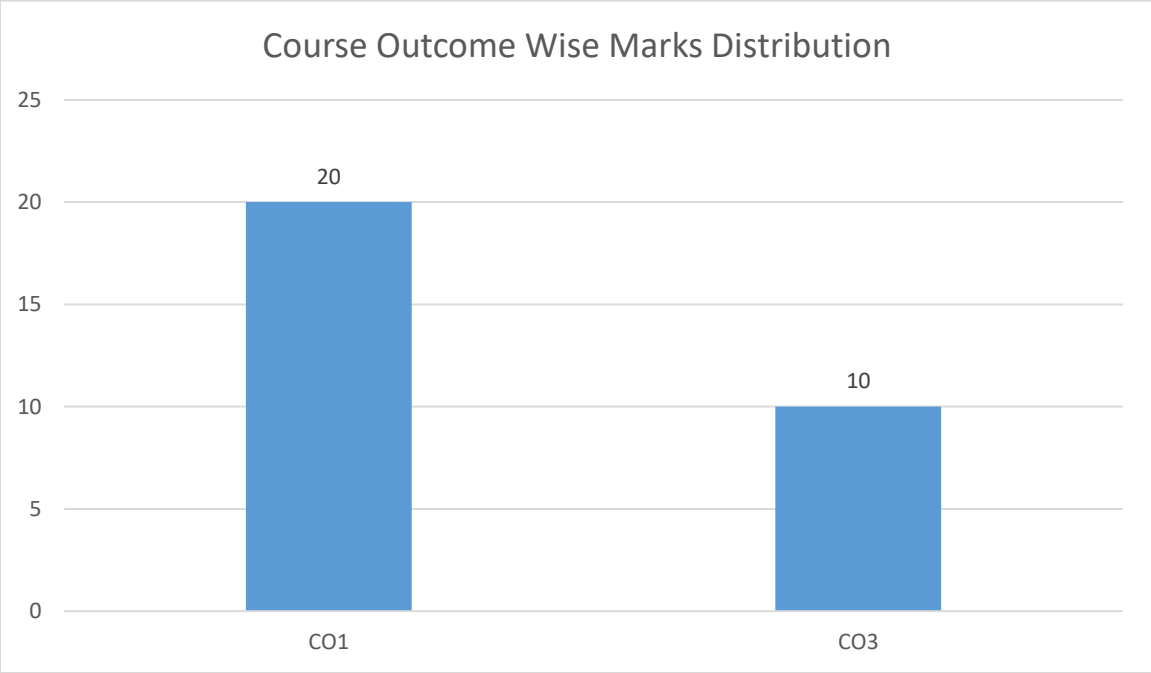
Note: All questions are mandatory. Attempt all the questions in sequence. Assume any missing data.

Q.No.	Questions	Marks	CO	BL
Q1(a)	State the difference between Key-value database and Document-oriented database with proper examples.	**5**	CO3	L2
Q1(b)	Define CAP theorem with respect to MongoDB.	**5**	CO3	L1
Q2.	Discuss all comparisons operators of MongoDB with proper examples.	**6**	CO1	L2

Q3.	Consider the collection named "student", which consists of the following documents.
	{
	_id: "S01",
	name: "John",
	subjects: ["SQL", "NoSQL", "Java"] ,
	scores: [67, 78, 89]
	age: 22,
	address:{
	city: "Patiala",
	state: "Punjab"
	}
	}
	{
	_id: "S02",
	name: "Mark",
	subjects: ["Javascript", "Python", "Java"] ,
	scores: [88, 64, 79],
	age: 21,
	address:{
	city: "Kolkata",
	state: "WB"
	}
	}
	{
	_id: "S03",
	name: "Bob",
	subjects: ["SQL", "Python", "Java"] ,
	scores: [78, 68, 69],
	age: 23,
	address:{
	city: "Chennai",
	state: "TN"
	}
	}
			CO1	L3

3(a)	Write a MongoDB query to add three subjects "DS", "DBMS", "SE" in the subject list of John. It should be noted that the newly added three subjects should appear as initial elements of the subjects array.	**4**	CO1	L3
3(b)	Write a MongoDB query to update the scores of all student in such a way that it should contain only highest three marks.	**4**	CO1	L3
3(c)	Find all the documents where the subjects list of students contain only "SQL" and "NoSQL" in the specified order.	**2**	CO1	L3
3(d)	Find all documents where the scores of student contain at least one element between 80 and 90.	**2**	CO1	L3
3(e)	Write a MongodB query to increment the age of Mark by 2 years.	**2**	CO1	L3


Bloom's Level Wise Marks Distribution
5
11
14
Level 1
Level 2
Level 3


Course Outcome Wise Marks Distribution
25
20
15
10
5
0
20
10
CO1
CO3